

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION II
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/03

Paper 3 Applications Paper

25 September 2013

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

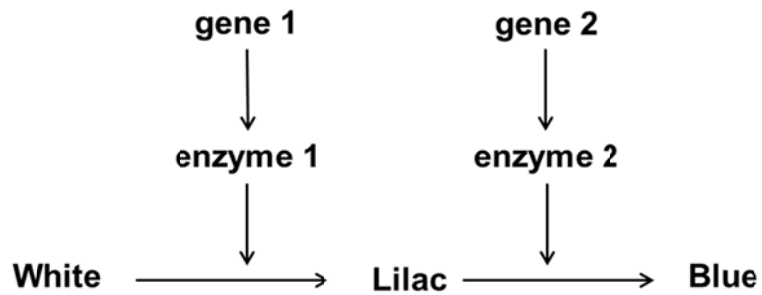
For Examiner's Use	
1	15
2	15
3	10
4	12
5	20
Total	72
Percentage	
Grade	

This document consists of **16** printed pages.



Section AAnswer **all** questions.

- 1 In flax plants, the flowers are white, lilac or blue. Fig.1 shows the pathway by which the flower cells produce coloured pigments.

**Fig. 1**

- (a) A single base deletion occurs in gene 1.

Describe and explain the effect of this mutation on the colour of flax plant flowers.

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[4]

- ## Table 1

!sis	s
	
	
	
	

- [5]

- (c) Genomic libraries and cDNA libraries can be created for the flax plant genome.

State **three** differences between the two gene libraries.

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..... [3]

[Total: 15]

- 2 Diabetes mellitus is a human disease condition in which the patients do not produce sufficient insulin. A way to treat diabetic patients is to inject purified recombinant human insulin, produced using genetically engineered bacteria, into their bloodstream.

Insulin is a relatively small protein comprising two polypeptides, one of 21 amino acids (the A chain) and one of 30 amino acids (the B chain). Trinucleotides representing all the codons are synthesized and joined together to form two artificial genes for the respective chains. Fig. 2.1 shows the two artificial genes, including the positions of the recognition sequences of two restriction enzymes, *Pst*I and *Eco*RI, and distances (in base pairs) between the sites.

Each artificial gene is cloned into a separate bacterial plasmid pUC21, using the restriction enzyme *Hind*III, and placed under the control of a strong *lac* promoter. Fig. 2.2 shows the plasmid map of pUC21. It includes the positions of the restriction sites for the three restriction enzymes, *Pst*I, *Eco*RI and *Hind*III, and distances (in base pairs) between the sites. The shaded regions represent two marker genes, *Amp*^R and *lac* Z.

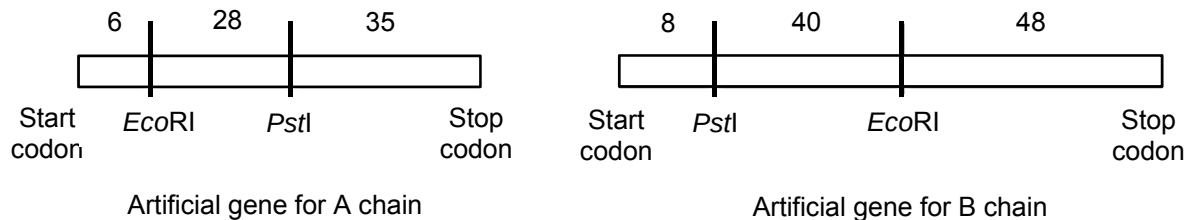


Fig. 2.1

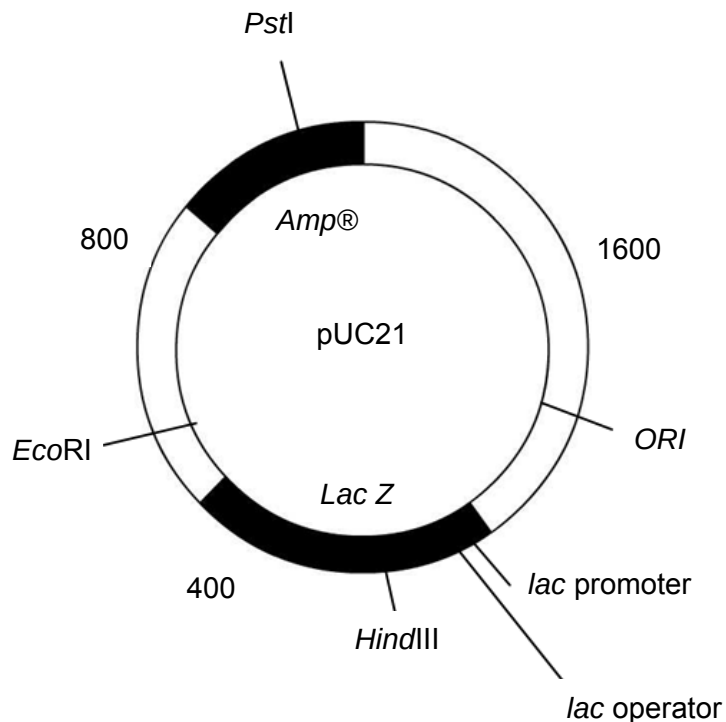


Fig. 2.2

Both recombinant plasmids containing the two artificial genes are transformed separately into *E. coli* cells. The bacterial cells from the transformation process are grown on an agar plate containing ampicillin and X-Gal.

- (a) State the substance that must be added to induce the expression of the artificial genes in *E. coli*.

..... [1]

- (b) Explain what is meant by a *marker gene*.

..... [1]

- (c) Heat shock treatment is applied to *E.coli* cells to render them competent to take up recombinant plasmids.

Describe **three** ways in which this process of artificial transformation of the recombinant plasmid into bacterial cells differs from natural transformation.

..... [3]

- (d) Describe and explain the procedure to select for bacterial cells containing the recombinant plasmid.

..... [4]

Once the transformed bacterial colonies have been identified, nucleic acid hybridization is carried out to further identify the colonies which contained the artificial genes.

The plasmid DNA from these colonies is then isolated and subjected to digestion by a particular restriction enzyme to determine which colonies have the artificial genes inserted in the correct orientation.

A student who is unsure of which particular restriction enzyme to use carried out digestion of all the four possible recombinant plasmids numbered A1, A2, B1 and B2 using either enzyme *EcoRI* or *PstI*.

Table 2 shows all the results obtained.

Table 2

	Digestion of plasmid DNA with artificial gene for A chain				Digestion of plasmid DNA with artificial gene for B chain			
Plasmid	A1	A2	A1	A2	B1	B2	B1	B2
Length of fragments (bp)	406	463	1234	1235	1288	1208	448	448
	2463	2406	1635	1634	1608	1688	2448	2448
Restriction enzyme used								

- (e) With reference to Fig. 2.1 and Fig. 2.2, complete Table 2 by identifying the particular restriction enzyme used for digestion of the respective plasmid DNA. [2]
- (f) With reference to Fig. 2.1, Fig. 2.2 and Table 2, state the appropriate restriction enzyme that has to be used for the digestion of the respective plasmid DNA in order to determine which has the artificial genes inserted in the correct orientation.
- (i) Restriction enzyme to digest plasmid DNA with artificial gene for A chain
- (ii) Restriction enzyme to digest plasmid DNA with artificial gene for B chain

[2]

(g) Provide an explanation for your answer in (f).

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..... [2]

[Total: 15]

- 3 Duchenne muscular dystrophy (DMD) is a common lethal X-linked human genetic disease caused by the absence of the protein dystrophin in muscle fibres.

Possible treatments included the transplanting of normal cells from genetically compatible donors or gene therapy which involves incorporating recombinant DNA with part of the normal allele of the gene for dystrophin into a muscle.

(a) Explain briefly what is meant by *genetically compatible donors*.

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..... [2]

The dystrophin gene was inserted into an adenovirus and injected into a strain of mice with the symptoms of muscular dystrophy.

Eight mice with DMD were injected once in a muscle in one hind limb with different doses of the modified virus 7 days after birth. The mice were then killed at the age of 2 to 25 weeks and sections of the injected muscle and uninjected muscle from the other hind limb were obtained.

These tissues were then treated with a fluorescent antibody specific to dystrophin to show gene expression and the results are shown in Table 3.

Table 3

Mouse	Virus dose/ particles*10 ⁹ cm ⁻³	Death/weeks after birth	Increase in muscles fibres with dystrophin/%
1	1.0	2	6
2	2.0	3	8
3	2.5	5	10
4	3.0	8	22
5	4.0	10	30
6	5.0	14	50
7	5.0	18	35
8	5.0	25	15

(b) Describe and explain how the adenovirus delivers the dystrophin gene.

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..... [3]

(c) With reference to Table 3,

- (i) suggest why uninjected muscle from each mouse was also treated with fluorescent antibody specific to dystrophin.

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..... [1]

- (ii) describe the effect of the injected dosage of virus on the percentage of muscle fibres producing dystrophin.

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..... [2]

- (iii) explain why this gene therapy needs to be repeated.

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..... [2]

[Total: 10]

4 Planning question

Malt is germinated cereal grains which have been dried in a process known as malting. Malted barley contain amylases which digest starch into maltose units by cleaving the α -1,4-glycosidic bonds.

The digestion of starch is an exothermic process and releases energy. This energy released can be detected by the largest increase in temperature of the reaction mixture over a period of time.

Using this information and your own knowledge, design an experiment to determine the concentration of active amylase in malted barley samples that have undergone different heat treatments.

You must use:

- 1.0 moldm⁻³ starch solution
- 10% amylase solution
- Malted barley which have heat treated for 10 minutes at 40.0 °C, labelled **M1**
- Malted barley which have been heat treated for 10 minutes at 80.0°C, labelled **M2**
- distilled water
- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes, pipette fillers
- timer e.g. stopwatch or stopclock
- mercury thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms, include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

[illegible]

[illegible]

[illegible]

[illegible]

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b), etc., as indicated in the question.

5 (a) Explain the normal functions of stem cells in a living organism. [6]

(b) Discuss the advantages and limitations of plant cloning. [8]

(c) Comment on the statement “GM crops can make a relevant contribution to solving health problems in developing countries.” [6]

[Total: 20]

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